

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 04

Sequential Circuit



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Recap of Previous Lecture



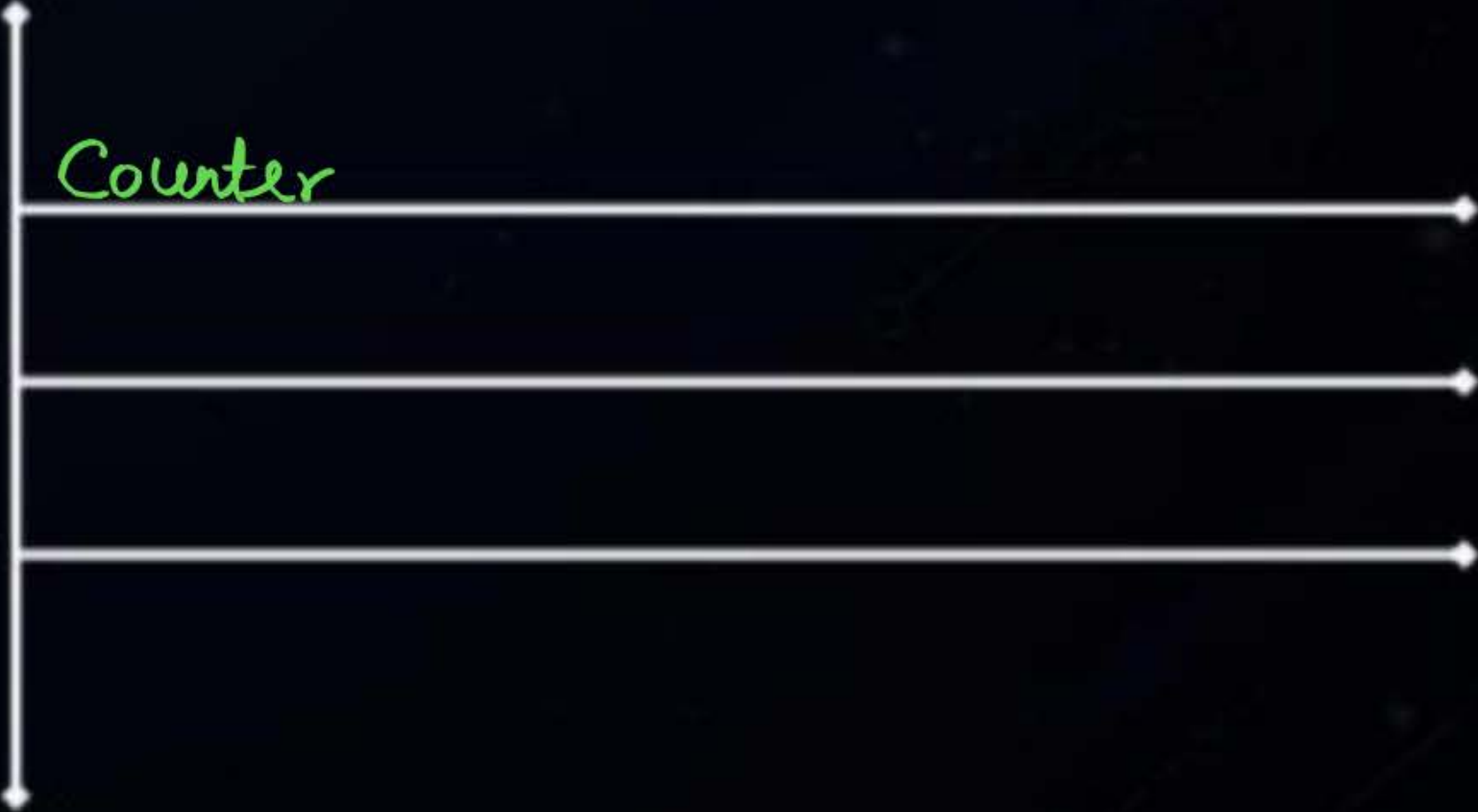
Race Around condition

Counter



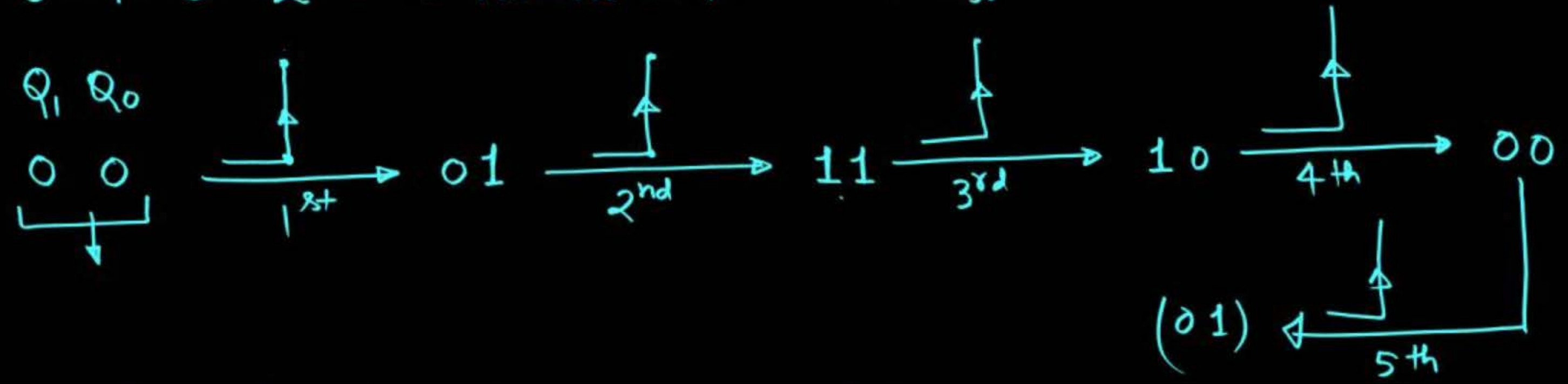
Topics to be Covered

Counter



• 0-0-1-1-2-2-2-3-3-3-3 → MOD no. = 11

• 0-1-3-2 → MOD no. = 4 → 2ff

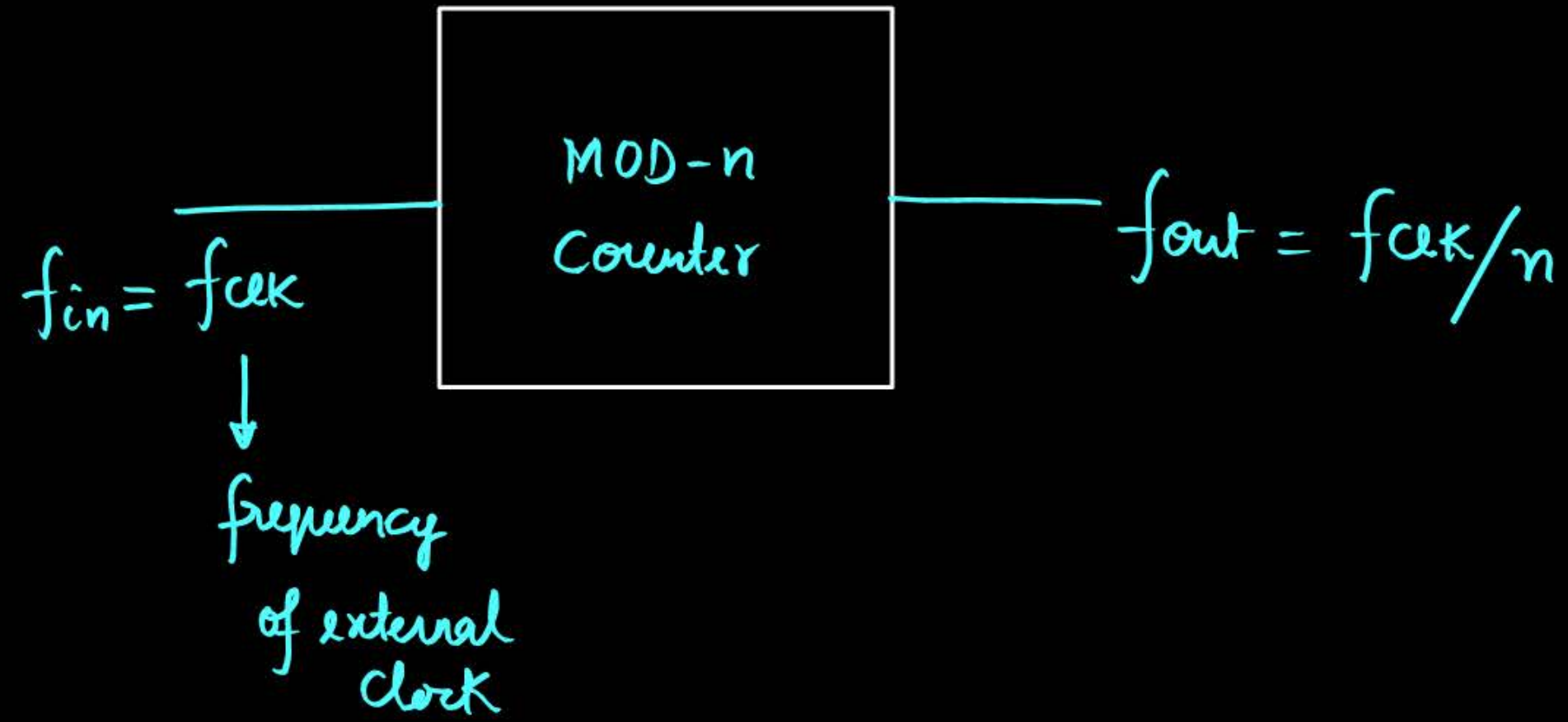


- Note: After application of MOD no. of clock pulses counter will again be at the starting state.
- After application of integer multiple of MOD no. of clock pulses counter will again be at starting state.

if starting state is $(3)_{10}$ then after 263 clock pulses, what will be the state of the counter $(1)_{10}$

$$(3)_{10} \xrightarrow{4\text{CLK}} (3)_{10} \xrightarrow{4\text{CLK}} (3)_{10}$$

$$(3)_{10} \xrightarrow[\text{260}]{65 \times 4} (3)_{10} \xrightarrow{261} 2 \xrightarrow{262} 0 \xrightarrow{263} 1$$



• 0-1-3-2

$f_{clk}/4$

Q_1, Q_0

0 0

0 1

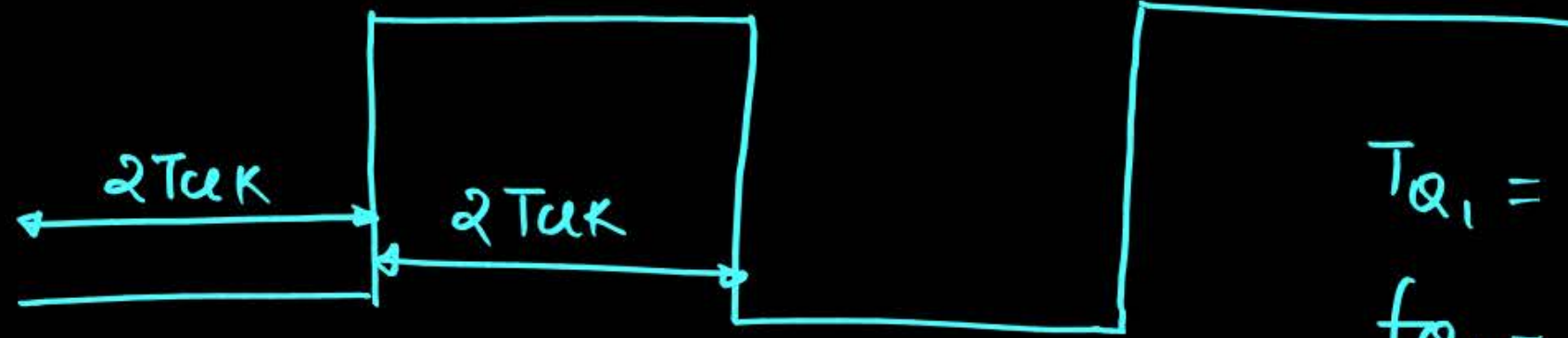
1 1

1 0

0 0

MOD no. = 4

$Q_1 \rightarrow T_{ON} = 2T_{clk}, T_{off} = 2T_{clk}$



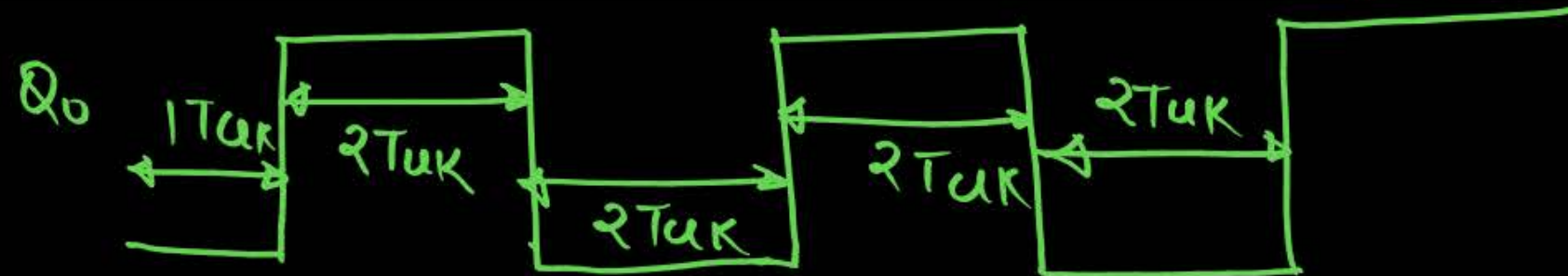
$$T_{Q_1} = T_{ON} + T_{off} = 4T_{clk}$$

$$f_{Q_1} = \frac{1}{T_{Q_1}} = \frac{1}{4T_{clk}} = \frac{f_{clk}}{4}$$

$Q_0 \rightarrow T_{ON} = 2T_{clk}, T_{off} = 2T_{clk}$

$$T_{Q_0} = T_{ON} + T_{off} = 4T_{clk}$$

$$f_{Q_0} = f_{clk}/4$$



- 0-1-4-3-2-5 MOD no = 6

	Q_2	Q_1	Q_0
	0	0	0
	0	0	1
	1	0	0
	0	1	1
	0	1	0
	1	0	1

$$Q_0 \rightarrow T_{off} = 1T_{clk}, T_{on} = 1T_{clk}$$

$$T_{Q_0} = 2T_{clk} \quad f_{Q_0} = f_{clk}/2$$

$$Q_1 \rightarrow T_{off} = 4T_{clk}, T_{on} = 2T_{clk}$$

$$T_{Q_1} = 6T_{clk} \rightarrow f_{Q_1} = f_{clk}/6$$

$$Q_2 \rightarrow T_{off} = 2T_{clk}, T_{on} = 1T_{clk}$$

$$T_{Q_2} = 3T_{clk}, f_{Q_2} = f_{clk}/3$$



• 0-1-4-2

MOD no = 4

↳ 3 ff

	Q_2	Q_1	Q_0
	0	0	0
└─┐	0	0	1
└─┐	1	0	0
└─┐	0	1	0

$$Q_0 \rightarrow T_{ON} = 1T_{CLK}, T_{OFF} = 3T_{CLK}$$

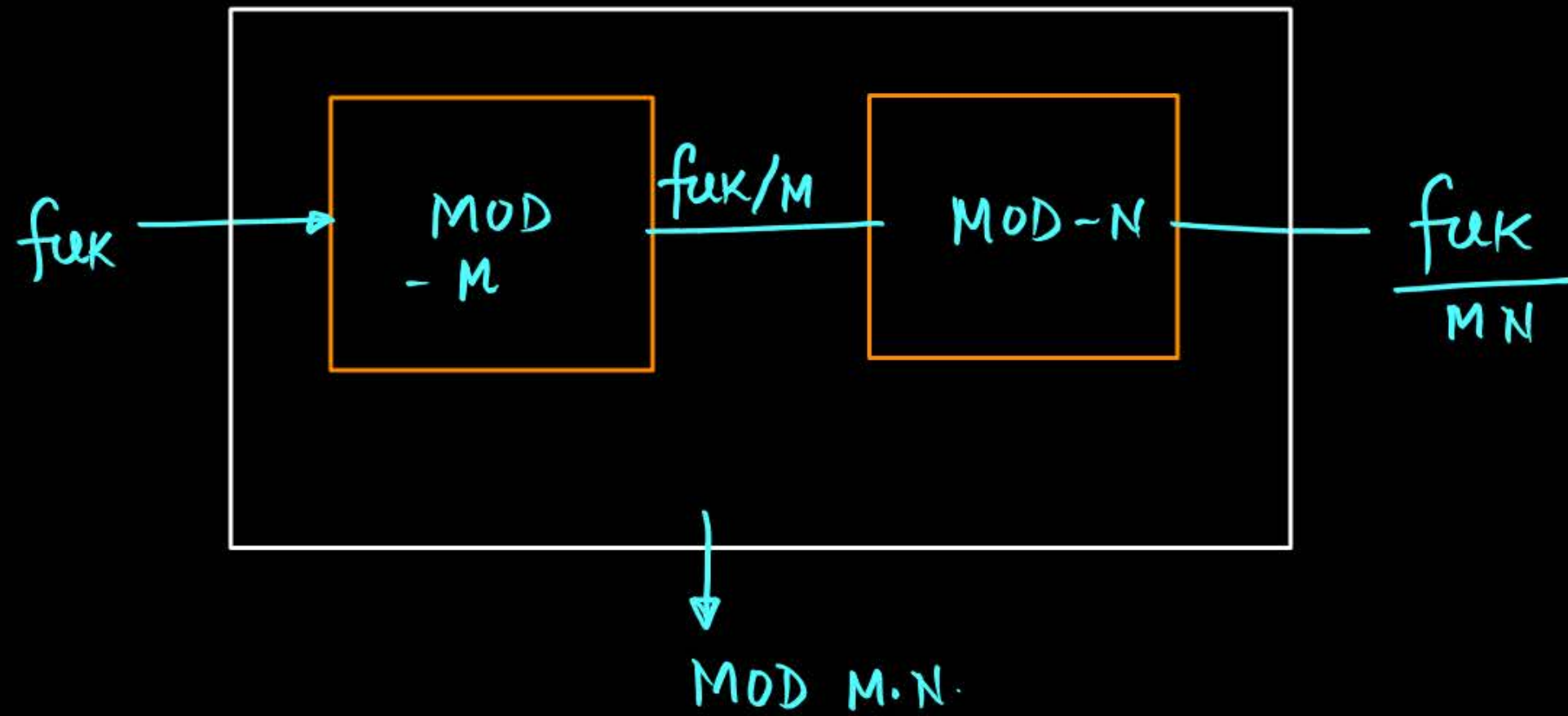
$$T_{Q_0} = 4T_{CLK}, f_{Q_0} = \frac{1}{4}f_{CLK}$$

$$Q_1 \rightarrow T_{ON} = 1T_{CLK}, T_{OFF} = 3T_{CLK}$$

$$T_{Q_1} = 4T_{CLK}, f_{Q_1} = f_{CLK}/4$$

$$Q_2 \rightarrow T_{ON} = 1T_{CLK}, T_{OFF} = 3T_{CLK}$$

$$T_{Q_2} = 4T_{CLK}, f_{Q_2} = f_{CLK}/4$$



Q. We have a MOD-4 counter and MOD-10 counter cascaded, then what is the resultant MOD no 40.

[Types of Counter]



Asynchronous Counter

- Here different ffs are driven by different clocks.

- Only fix sequence is possible

up sequence down sequence

- eg. up counting sequence
down counting sequence
- It is relatively slower.

Synchronous Counter

- Here all the ffs are driven by same clock.

- All possible sequences can be designed.

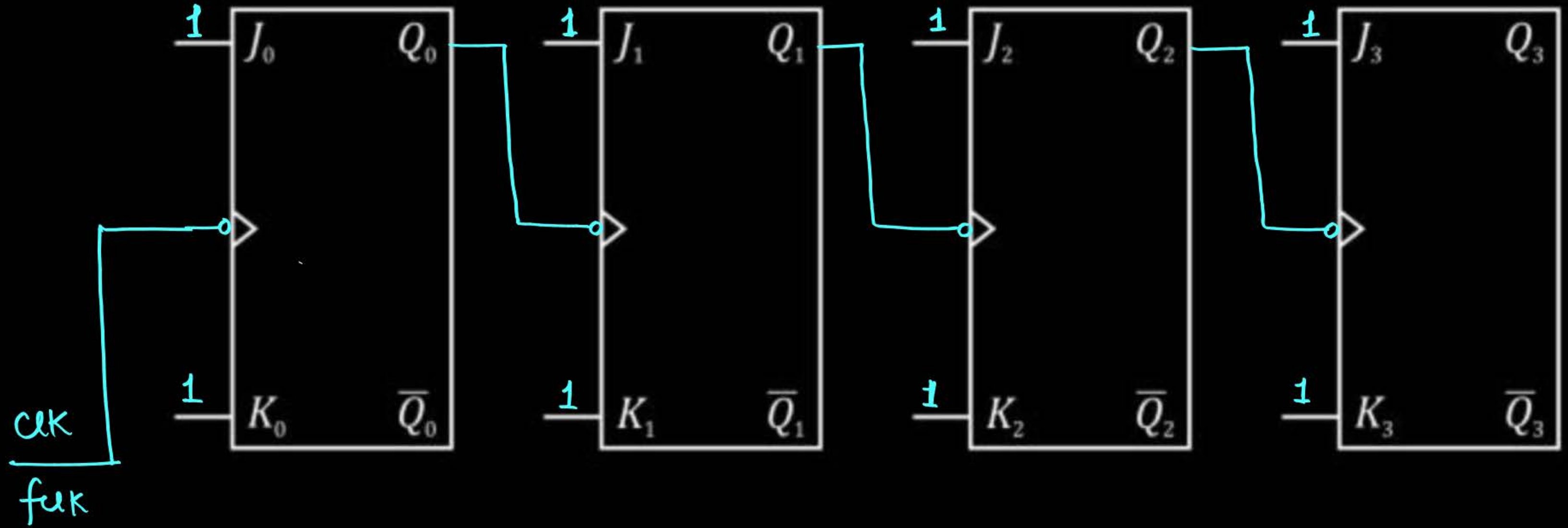
eg. Ring Counter, Johnson Counter etc
Any sequence.

- It is relatively faster.

[4-bit Asynchronous Counter]



- Circuit:



- Working:

- $Q_0 \longrightarrow$ toggles in every clock cycle.
- $Q_1 \longrightarrow$ toggles when Q_0 take transition from (1-0).
- $Q_2 \longrightarrow$ toggles when Q_1 take transition from (1-0)
- $Q_3 \longrightarrow$ toggles when Q_2 take transition from (1-0)



MOD-16
Counter.

	Q_3	Q_2	Q_1	Q_0
Starting state	0	0	0	0
1 st clock	0	0	0	1
2 nd clock	0	0	1	0
3 rd clock	0	0	1	1
4 th clock	0	1	0	0
5 th clock	0	1	0	1
6 th clock	0	1	1	0
7 th clock	0	1	1	1
8 th clock	1	0	0	0
9 th clock	1	0	0	1

$Q_0 \rightarrow T_{off} = 1T_{clk}, T_{on} = 1T_{clk}$
 $T_{Q_0} = 2T_{clk} \rightarrow f_{Q_0} = f_{clk}/2$

$Q_1 \rightarrow T_{off} = 2T_{clk}, T_{on} = 2T_{clk}$
 $T_{Q_1} = 4T_{clk}, f_{Q_1} = \frac{f_{clk}}{4}$

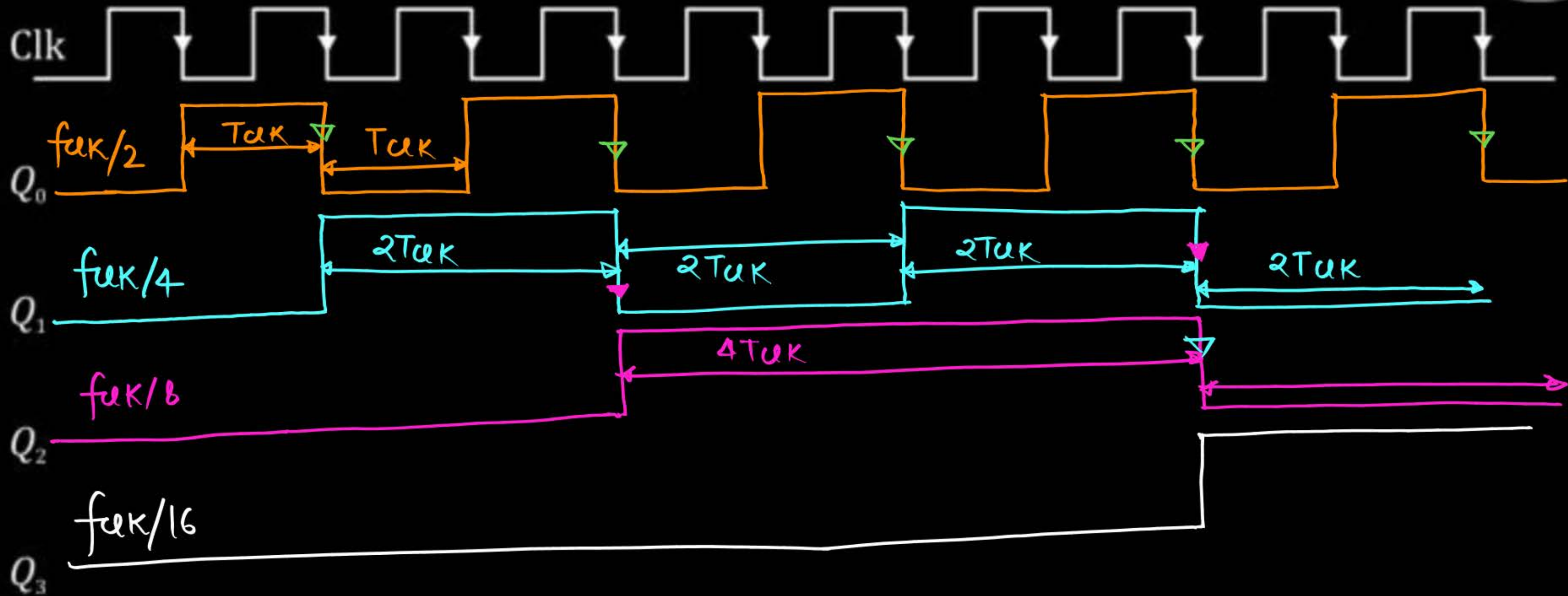
$Q_2 \rightarrow T_{off} = 4T_{clk}, T_{on} = 4T_{clk}$
 $T_{Q_2} = 8T_{clk}, f_{Q_2} = \frac{f_{clk}}{8}$

$Q_3 \rightarrow T_{off} = 8T_{clk}, T_{on} = 8T_{clk}$
 $T_{Q_3} = 16T_{clk}, f_{Q_3} = \frac{f_{clk}}{16}$



10 th	1	0	1	0
11 th	1	0	1	1
12 th	1	1	0	0
13 th	1	1	0	1
14 th	1	1	1	0
15 th	1	1	1	1
16 th clock	0	0	0	0

- Clock Diagram :





Topic : 2 Min Summary

→ Asynchronous Counter.

Thank you

GW
Soldiers !

